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Divergent Disinhibition Profiles Linked to Externalizing Psychopathology

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Abstract

Disinhibited individuals show heterogeneity in their neuropsychosocial characteristics. To better understand this variability, we used a person-centered approach to identify transdiagnostic profiles of disinhibition based on a range of personality and neurocognitive indicators. A community sample of adults (N = 363; ages 18–55; 50.1% female; data collected 2019–2024), enriched for externalizing disorders, completed a multilevel assessment encompassing clinical symptoms, personality traits, neuropsychological task performance, and trauma history. Latent profile analysis of disinhibition-related indicators identified three distinct profiles that differed in both the severity and configuration of disinhibitory features. Two profiles exhibited elevated disinhibition: the *Neurocognitive Disinhibition* profile (n = 72) was characterized by low cognitive and executive functioning performance but average impulsive personality traits, whereas the *Personality Disinhibition* profile (n = 103) was marked by high levels of trait impulsivity and intact cognitive-executive functioning. A third profile, *Low Disinhibition* (n = 188), was characterized by relatively low impulsivity and strong executive functioning. The *Neurocognitive* and *Personality Disinhibition* profiles showed higher levels of externalizing psychopathology and self- and other-directed violence compared with the *Low Disinhibition* profile. In addition, the *Neurocognitive Disinhibition* profile demonstrated greater specificity for externalizing psychopathology and a more extensive history of trauma exposure, whereas the *Personality Disinhibition* profile was characterized by elevations across both externalizing and internalizing psychopathology. Together, these findings indicate that symptom-agnostic, person-centered classification approaches can delineate meaningful subtypes of disinhibition that differ across personality, neurocognitive, and environmental risk domains.

General Scientific Summary:

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Author Contributions

Conceptualization: Sadeh; Funding Acquisition: Sadeh; Formal Analysis: Sadeh, Stumps; Methodology: Sadeh, Spielberg; Writing - Original Draft Preparation: Sadeh, Stumps; Writing - Review & Editing: Sadeh, Stumps, Spielberg.

All authors participated in writing the paper and approved the final version for submission.

Conflicts of Interest

The author(s) declare that there were no conflicts of interest with respect to the authorship or the publication of this article.

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People who struggle with impulsive and risky behaviors may differ in important ways that are not necessarily captured by traditional diagnoses. This study identified three distinct subgroups of individuals based on measures of impulsive personality traits and related cognitive skills. Results offer new insight into the different combinations of factors that can lead to impulsive behavior and highlight the value of looking beyond symptoms to better understand and treat these individuals.

Keywords

impulsive personality traits; delay discounting; executive function; inhibition; trauma

Externalizing psychopathology encompasses a range of serious public health concerns, including violence, substance use, criminal behavior, and other forms of risk-taking, and is operationalized as a spectrum of related disorders that includes antisocial personality disorder, conduct disorder, alcohol and substance use disorders, attention deficit hyperactivity disorder (ADHD), and related impulse-control conditions (American Psychiatric Association, 2013; Krueger et al., 2002; Krueger et al., 2021). These problems are often chronic, treatment-resistant, and associated with markedly reduced life expectancy (e.g., Abdul-Rahman et al., 2018; Krasnova et al., 2019). Traditional diagnostic approaches have limitations in capturing the complexity of externalizing disorders, as they often overlook high rates of comorbidity among externalizing disorders (Beauchaine et al., 2017; Krueger et al., 2002), shared transdiagnostic mechanisms across these disorders (Krueger & Eaton, 2015), and etiological heterogeneity among individuals with similar symptom presentations (Driessen et al., 2018; Fanti & Kimonis, 2017; Sadeh et al., 2020). Because surface-level symptoms often do not reflect the core mechanisms maintaining psychopathology, there is growing interest in identifying the causal processes that confer vulnerability to externalizing disorders (Krueger & Markon, 2006; Young et al., 2009).

Transdiagnostic frameworks offer a promising approach for elucidating the diverse and multifactorial etiology of disinhibited behaviors that characterize externalizing psychopathology (Kotov et al., 2017; Lynch et al., 2021). One such core transdiagnostic mechanism is disinhibition, which functions as a central contributor to externalizing pathology (Krueger et al., 2002). An unresolved question in the study of disinhibition is whether it primarily reflects a single, broad liability that underlies externalizing psychopathology or whether similar behavioral manifestations can arise from multiple, distinct pathways, such as abnormalities in cognitive or motor functions that support inhibitory control and disinhibition stemming from personality traits (e.g., Nigg, 2000; Nigg, 2017). Thus, individuals may exhibit comparable levels of disinhibition despite substantial differences in etiology. To better understand variability in the causes of disinhibition and their importance for explaining psychopathology, this study used a community sample of adults to (i) identify distinct profiles using broad, cross-cutting disinhibition-related cognitive and personality indicators and (ii) examine how these profiles differ in clinical symptoms and trauma exposure.

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Personality Trait and Cognitive Mechanisms of Disinhibition

One extensively studied manifestation of disinhibition is trait-level impulsivity, or the tendency to act rashly, exhibit poor restraint, and behave without considering consequences (Berg et al., 2015; Kotov et al., 2010; Whiteside & Lynam, 2001). Within multidimensional models of trait impulsivity, impulsive urgency (i.e., frequent emotion-driven impulsivity), sensation seeking (i.e., drive for novel or stimulating experiences), and low conscientiousness (i.e., a tendency to act without thinking or planning) have all been linked to a range of externalizing outcomes (Berg et al., 2015; Kotov et al., 2010; Whiteside & Lynam, 2001). These traits map directly onto the disinhibited externalizing spectrum described in the HiTOP model, which conceptualizes disinhibition as a broad, trait-based liability characterized by impulsivity, irresponsibility, and risk-taking (Kotov et al., 2017; Krueger et al., 2021). Longitudinal and neurobiological studies further demonstrate that trait impulsivity is highly heritable, serves as a developmental precursor to externalizing disorders, and is associated with disruptions in brain systems responsible for inhibition and other executive functions (Beauchaine et al., 2017; Iacono et al., 2008). Thus, dimensions of trait impulsivity represent robust dispositional correlates of externalizing psychopathology.

In parallel, extensive research has linked disturbances in multiple cognitive systems to disinhibited behavior and externalizing disorders. Inhibitory control, or the ability to suppress automatic responses and dominant urges (Friedman & Miyake, 2004), is a core executive function that is particularly susceptible to disruption in contexts that tax cognitive resources, activate reward circuitry, or heighten emotional arousal. Such disruptions impair self-regulation and increase the likelihood of impulsive or maladaptive behavior (Bounoua et al., 2025; Hartikainen et al., 2012; Hofmann et al., 2012; Volkow et al., 2012). Working memory, the capacity to temporarily hold and manipulate information in service of goal-directed behavior, similarly supports self-regulation by maintaining task goals in the face of distraction (Bickel et al., 2014). Deficits in working memory prospectively predict the onset of substance use disorders (Khurana et al., 2017) and other externalizing outcomes (Hofmann et al., 2012). Finally, delay discounting, defined as the tendency to devalue rewards as a function of delay, reflects another facet of cognitive control (Odum et al., 2020). The ability to forego smaller, immediate rewards in favor of larger, delayed ones relies on inhibitory processes and future-oriented decision-making, and steeper delay discounting is consistently associated with impulsivity, addiction, and externalizing psychopathology (Odum et al., 2020; Volkow et al., 2012).

Although these personality and cognitive indicators relate to disinhibition, they capture distinct but complementary processes: trait-like tendencies toward rash action versus the cognitive functions that support the regulation of those tendencies. Notably, whereas the personality indicators align directly with dimensional models, including HiTOP's disinhibited externalizing spectrum (Krueger et al., 2021), the cognitive indicators are not as clearly specified in such models. Distinguishing between these levels of analysis may provide a framework for examining how trait dispositions and executive control processes jointly and independently shape heterogeneous pathways to externalizing psychopathology.

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A Person-Centered Approach to Disinhibition

Although disinhibition is often treated as a unitary construct, research increasingly shows that individuals with disinhibition-related problems exhibit diverse neuropsychosocial vulnerabilities (Martz et al., 2021; Ricard et al., 2024; Sadeh et al., 2021; Skeem et al., 2007). Most studies to date have used variable-centered methods, such as regression with interaction terms, to examine how combinations of risk factors contribute to these outcomes. Although useful for testing simple models, this approach becomes impractical and statistically underpowered when analyzing larger sets of variables, due to the exponential growth in interaction terms. These variable-centered approaches leverage variance *between individuals* on indicator variables to identify relationships among variables, whereas person-centered methods, such as latent profile analysis (LPA), leverage variance *within individuals* across indicators to identify subgroups of individuals who share similar patterns of risk factors. This latter approach allows researchers to capture how multiple influences co-occur and interact within people, yielding complementary insight into the complex pathways underlying disinhibition.

A growing body of research supports the utility of person-centered approaches for uncovering diverse neuropsychosocial profiles among disinhibited individuals. For example, a recent longitudinal study of adolescents identified nine distinct profiles based on a broad range of neurocognitive metrics (e.g., executive function, general cognition, learning and memory), and demonstrated that the profiles with below-average performance on most neurocognitive measures exhibited the highest levels of externalizing behaviors (Paskewitz et al., 2024). Another recent study used LPA to uncover risk profiles for antisocial behavior using personality, family, and broader environmental indicators of risk for these behaviors (Ricard et al., 2024). This study identified five profiles that varied in combinations of risk factors and found that the profile with elevated risk across the personality, family, and neighborhood levels of analysis showed the strongest association with antisocial personality disorder. Together, these findings underscore the value of person-centered approaches in revealing meaningful heterogeneity in disinhibition-related risk for externalizing psychopathology. Nonetheless, further research is needed to clarify the diverse etiological pathways underlying externalizing psychopathology. To our knowledge, no prior studies have investigated whether indicators of disinhibition spanning personality traits and cognitive functioning combine in unique ways to confer risk for externalizing behaviors. Addressing this gap is a central aim of the present study.

Current Study

Building on this growing literature, the present study employed LPA to identify distinct disinhibition profiles using seven indicators spanning personality traits and cognitive functioning. Participants were recruited from the community to capture dimensional variation in disinhibition across a broad range of severity that ranged from psychiatrically healthy individuals to those meeting criteria for lifetime psychopathology. Based on previous research (Paskewitz et al., 2024; Ricard et al., 2024; Sadeh et al., 2021; Skeem et al., 2007), we expected at least one profile to show elevated impulsive personality traits and low cognitive functioning, consistent with previous findings (Paskewitz et al., 2024;

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Ricard et al., 2024). We also expected to identify a low disinhibition profile characterized by relatively low levels of impulsive personality traits and intact cognitive regulatory functions (e.g., Paskewitz et al., 2024; Ricard et al., 2024).

To contextualize the findings in the broader literature, we also examined the external correlates of the latent profiles. Lifetime externalizing psychopathology was the primary external correlate of interest in this study. We hypothesized that disinhibition-based profiles would differ in lifetime externalizing diagnoses, with profiles characterized by higher impulsive traits and lower cognitive control exhibiting the highest rates of externalizing disorders. To provide a fuller picture of the clinical characteristics associated with each profile, we also examined rates of internalizing disorders and externalizing–internalizing comorbidity. These internalizing comparisons were included to contextualize the broader psychopathology profiles of each group; however, we did not have specific hypotheses regarding these associations.

Given their clinical relevance, we also examined violence perpetration and trauma history as external correlates of the disinhibition profiles. Extensive research indicates that externalizing psychopathology confers heightened risk for both self- and other-directed violence. More specifically, individuals with externalizing disorders are at elevated risk of engaging in self-harm behaviors, including suicide attempts and non-suicidal self-injury (NSSI) (Meszaros et al., 2017; Verona et al., 2004). Further, the perpetration of violence against others, such as reactive aggression and physical assaults, occurs at elevated rates in externalizing populations relative to the general population (Chow et al., 2024). Accordingly, we examined whether disinhibition profiles differed in their tendency to engage in self-directed and other-directed violent behavior, hypothesizing that profiles characterized by elevated trait and cognitive disinhibition would exhibit higher levels of self- and other-directed violence.

Second, a substantial body of work has identified trauma exposure, such as physical abuse, sexual assault, and exposure to community violence, as a significant risk factor for the development of disinhibition (Maxfield & Widom, 1996; Rogers et al., 2022). Importantly, prior research suggests that different forms of trauma may confer differential risk for externalizing outcomes, with interpersonal or assaultive traumas often showing stronger associations with impulsivity, aggression, and antisocial behavior than non-assaultive stressors (e.g., accidents, natural disasters) (Bynion et al., 2018; Heleniak & McLaughlin, 2020; Maxfield & Widom, 1996). Traumatic experiences are thought to increase vulnerability to impulsivity, aggression, and antisocial behavior through disruptions in emotion regulation, impairments in executive functioning, and heightened physiological reactivity to stress (De Bellis, 2001; McLaughlin et al., 2014). For some individuals, posttraumatic psychopathology may also manifest as increased sensation seeking or risk-taking behavior, potentially as a compensatory response to emotional numbing or as a consequence of heightened arousal and diminished inhibitory control under stress (Kerig, 2019). At the same time, longitudinal evidence suggests that pre-existing personality traits related to disinhibition (e.g., low constraint) may increase liability for subsequent exposure to stressful or potentially traumatic events (Haller & Chassin, 2012; Koffel et al., 2016; Sadeh et al., 2015), highlighting potential bidirectional pathways between trauma

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and disinhibition. Given that trauma exposure represents a theoretically and empirically meaningful external correlate of disinhibition, we examined lifetime trauma history across profiles, distinguishing between assaultive and non-assaultive trauma exposure. We expected profiles characterized by elevated trait and neurocognitive disinhibition to report greater exposure to assaultive trauma (e.g., Bynion et al., 2018).

Methods

Participants

Participants were recruited from the community through online platforms (e.g., Craigslist, Facebook) and physical flyers posted in public locations, including libraries, public transit stops, mental health facilities, and probation and parole offices. Recruitment materials described the study as focusing on decision-making, mental health, risky behavior, and life experiences in adults. Some advertisements included targeted language designed to recruit both psychiatrically healthy adults and individuals with varying levels of externalizing psychopathology, consistent with the study's goal of capturing dimensional variation in disinhibition and related behaviors across the full severity spectrum. For example, some flyers targeted healthy adults (e.g., *"Are you a healthy adult interested in participating in a study on risky behavior?"*), whereas others were intended to attract individuals with elevated impulsivity, risk-taking, and externalizing behaviors (e.g., *"Are you an impulsive person? Do you do things just for the thrill of it?"*; *"Have you ever gotten in trouble with the law or taken risks that led to problems?"*; *"Do you use drugs recreationally or have a history of alcohol or drug treatment?"*). Notably, 32.2% of the final sample did not meet criteria for any assessed externalizing or internalizing disorder, indicating that the sample included a substantial proportion of psychiatrically healthy individuals and supporting the study's dimensional analytic approach. Eligibility criteria included being between 18 and 55 years old and fluent in English.

The final sample consisted of 363 adults [182 women (50.1%), 181 men (49.9%)] aged 18–55 (*M/SD* = 32.72/10.58). The sample was diverse in terms of racial/ethnic identity, household income, and educational attainment. Approximately 46.1% identified with an ethno-racial group other than exclusively White Non-Hispanic. White was the most common self-reported race (61.9%), followed by Black (25.7%), Asian, (11.0%), and Other (1.4%), with 8.0% identifying as Hispanic. Annual household income ranged from $0 to $300,000, with a median income of $50,000. The most common level of educational attainment was a high school diploma equivalent or less (49.7%), with the remaining participants attaining an Associates or Bachelor's degree (33.2%) or a graduate degree (17.1%). Additionally, most of the sample reported a history of mental health treatment (59.1%, n = 214) and a substantial minority (41.6%, n = 151) reported prior involvement with the legal justice-system.

Procedures

Transparency and Openness. The data and syntax for the study are publicly available: https://sites.udel.edu/nsadeh/data-repository/ (Sadeh, 2026). We report all data exclusions, all manipulations, and all measures in the study. We did not conduct an *a priori* power

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analysis, as this study represents a secondary analysis of existing data collected as part of a larger project. The sample was drawn from a larger study, and the current analyses included all participants who completed the neuropsychological battery from the larger data pool. All participants completed a clinical interview, behavioral tasks, and questionnaire battery. Neuropsychological testing was administered by a licensed clinical psychologist or by trained research staff who demonstrated competency in standardized administration procedures under supervision. Informed consent was obtained before participation.

The University of Delaware's Institutional Review Board approved all protocols and procedures (Protocol #'s: 1073423–17). The study was not preregistered; however, all other procedures were carried out in accordance with the provisions of the World Medical Association Declaration of Helsinki.

Measures

Personality Traits. Participants completed the *UPPS-P Impulsive Behavior Scale* (Lynam et al., 2006), a 59-item self-report questionnaire designed to assess dimensions of impulsivity-related personality traits: negative urgency, positive urgency, lack of premeditation, lack of perseverance, and sensation seeking. The UPPS-P was selected over other impulsivity measures, because it distinguishes temperamental affect-driven impulsivity (negative and positive urgency) from cognitive and behavioral control deficits. Each item is rated on a 4-point scale from 1 (*Agree Strongly*) to 4 (*Disagree Strongly*). Consistent with prior work supporting a hierarchical three-factor model (Cyders & Smith, 2007), analyses were based on this higher-order structure: an Impulsive Urgency factor was derived by averaging scores from the negative and positive urgency subscales (Cronbach's alpha = .94; *M/SD* = 2.00/0.59), Sensation-Seeking was measured using its corresponding subscale (Cronbach's alpha = .87; *M/SD* = 2.58/0.63), and a Low Conscientiousness factor was computed by combining the premeditation and perseverance subscales (Cronbach's alpha = .88; *M/SD* = 1.86/0.45). Five participants were missing this measure.

Cognitive Functions. To index *inhibitory control*, participants completed the Color-Word Interference Test (Delis et al., 2001), and the scaled scores for the inhibition and inhibition/switching subtests were entered into analyses. The inhibition subtest requires participants to name the color of the ink in which words are printed while ignoring the word itself, and we averaged the scores on the reaction time and error rate indices (*M/SD* = 10.58/2.26). One participant was missing this variable. The inhibition/switching subtest requires participants to alternate between naming the ink color and reading the word based on specific visual cues, and we averaged the scores on the reaction time and error rate indices (*M/SD* = 10.21/2.27). There was no missingness on this variable.

To assess *working memory*, participants completed the Digit Span test from the Wechsler Adult Intelligence Scale-IV (WAIS-IV; Wechsler, 2008). The scaled total score was used in analyses (*M/SD* = 10.72/2.92). One participant was missing this variable.

Participants completed a *delay discounting* task to assess impulsive decision-making related to reward valuation. Delay discounting, also referred to as intertemporal choice, is a well-established approach for quantifying the preference for smaller immediate versus larger

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delayed rewards (Green & Myerson, 2004; Odum, 2011). In the task, participants made a series of choices between hypothetical monetary options that varied in size and delay with one option offering a smaller amount available sooner, and the other offering a larger amount available after a delay (Odum, 2011). Individual discounting tendencies were estimated using Mazur's (1987) hyperbolic function, which models how subjective value decreases as delay lengthens. The resulting discounting parameter (k-value) reflects the rate at which delayed rewards are devalued, with higher values indicating stronger preference for immediate rewards. Because k-values are typically positively skewed, they were natural log–transformed (ln k), such that higher scores represent greater discounting of future rewards (*M/SD* = −5.35/2.44). Twenty-five participants were missing this variable.

Lifetime Psychopathology Symptoms. Mental disorders were assessed with the *Structured Clinical Interview for DSM-5 Research Version* (*SCID-5-RV;* First et al., 2015). Using the *SCID-5-RV*, for the most severe lifetime episode (when applicable) for each of the following disorders: Alcohol Use, Substance Use (based on the drug selected as "most problematic" from cannabis, stimulants, opioids, sedatives, hallucinogens), Antisocial Personality, Borderline Personality, Attention Deficit Hyperactivity, Gambling, Major Depressive, Bipolar, Panic, Social Anxiety, and Generalized Anxiety. Research staff underwent extensive training on clinical interviewing, administration of the *SCID-5-RV*, and symptom ratings prior to conducting interviews. Independent secondary ratings were conducted on 90% of the interviews to assess reliability across interviewers and raters and indicated high reliability among the raters (Intraclass Correlation Coefficients ranged from = 0.93–0.98 for symptom totals). Lifetime diagnostic status was used for each disorder and for the externalizing and internalizing psychopathology spectra. For a lifetime externalizing disorder, participants needed to meet criteria for Alcohol or Substance Use Disorder, Antisocial or Borderline Personality Disorder, Attention Deficit Hyperactivity Disorder, or Gambling Disorder at some point in their lifetime. For a lifetime internalizing disorder, participants needed to meet criteria for Major Depressive Disorder, Bipolar I Disorder, Generalized Anxiety Disorder, Panic Disorder, or Social Anxiety Disorder. There was no missingness on these variables.

Self-Directed & Other-Directed Violence. Participants completed the self-report RISQ (Sadeh & Baskin-Sommers, 2017) and the interview-administered Life History of Aggression scale (LHA; Coccaro et al., 1997). Both measures had separate items assessing how many times individuals (i) engaged in non-suicidal self-injury (NSSI), (ii) attempted suicide, and (iii) perpetrated physical aggression against humans or animals over the course of their lives. Participants answered these questions using a six-point scale ranging from (0) "None" to (5) "More than 100 times" for the RISQ and from (0) "No events" to (5) "Too many times to count" on the LHA. For the current study, scores from both the RISQ and LHA were standardized (z-scored), averaged, and then Blom-transformed to create a composite index reflecting lifetime engagement in NSSI (Cronbach's alpha = .90; 22.1% endorsed 1 or more events), suicide attempts (Cronbach's alpha = .88; 11.5% endorsed 1 or more events), and physical aggression (Cronbach's alpha = .88; 63.4% endorsed 1 or more events).

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Trauma Exposure. Participants' histories of traumatic event exposure were assessed using the *Detailed Trauma Screen from the SCID-5-RV Stressor-Related Disorders* module (First et al., 2015) for 12 potentially traumatic events. Lifetime prevalence rates for the events assessed showed elevated rates of trauma exposure in the sample: domestic violence (39.8%), threatened with a weapon (37.0%), active war zone (2.2%), physical assault (29.3%), physical abuse (48.7%), sexual abuse (30.7%), terrifying medical events (16.0%), natural disaster (11.0%), serious accident (41.6%), death or serious injury to stranger (35.3%) or someone close to them (44.2%), and terrifying events at work (10.8%). *Assaultive trauma* was defined as direct exposure to threatened or actual physical violence perpetrated by another individual (e.g., domestic violence, being threatened with a weapon, exposure to an active war zone, or physical assault, sexual abuse; Cisler et al., 2012; Sadeh et al., 2015). *Non-assaultive trauma* was defined as exposure to non-interpersonal events, such as accidents, natural disasters, or terrifying medical procedures. Separate variables were created to index the total number of assaultive (*Min/Max* = 0/7; *M/SD* = 2.65/2.15) and non-assaultive (*Min/Max* = 0/4; *M/SD* = 0.79/0.85) trauma types endorsed across the lifespan. There were no missing data for these variables.

Data Analysis

First, we used latent variable mixture modeling in Mplus to identify homogeneous subgroups of individuals based on seven observed indicators reflecting trait and cognitive variables. Latent profile analysis (LPA) is a specific form of mixture modeling that uses continuous indicators to identify profiles of individuals with similar response patterns. All observed variables were standardized (z-scored) prior to analysis to facilitate interpretation across measures with different scales. Models were estimated using the robust maximum likelihood (MLR) estimator, which employs full-information maximum likelihood to handle missing data and is robust to non-normality in the observed variables. Models were estimated with 5,000 random sets of starting values, with the 100 best solutions retained for final optimization. Indicator variances were held equal across classes and covariances among indicators were fixed to zero, consistent with standard LPA assumptions (Nylund et al., 2007). For likelihood ratio tests (Lo-Rubin and Bootstrap LRTs), 40 random starts with 8 final optimizations were used. The final loglikelihood value was replicated across multiple random start sets, indicating that a stable global solution was likely achieved.

Recent simulation studies indicate that samples of our size (300–400) are adequate for identifying multiple latent profiles under conditions of strong class separation, particularly when entropy and posterior classification probabilities are high (Dalmaijer et al., 2023; Dziak et al., 2014). Model selection was guided by multiple fit indices, including the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), as well as entropy values reflecting class separation. We also used the Lo-Mendell-Rubin adjusted likelihood ratio test (LMR-A) and the bootstrapped likelihood ratio test (BLRT) to compare each model to one with one fewer class. The number of profiles was increased iteratively until these indices indicated that additional classes no longer improved model fit, and solutions with classes comprising less than 10% of the sample were excluded. Model estimation used random sets of starting values to ensure identification of the global solution. To characterize the emergent profiles, we examined differences between the profiles in the trait

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and cognitive disinhibition indicators used in the LPA analysis. These comparisons were conducted in SPSS (v29.0) using one-way ANOVAs with Games-Howell post hoc tests, which adjust for unequal variances and sample sizes.

To examine external correlates of the identified profiles while accounting for classification uncertainty, we applied the manual version of the Bolck-Croon-Hagenaars (BCH) three-step method (Asparouhov & Muthén, 2014). This approach retains the probabilistic class structure from the LPA and provides unbiased estimates of mean differences across profiles on external variables by incorporating classification error into the model. In Step 1, the optimal profile solution was identified, and posterior probabilities of class membership were saved. In Step 2, these probabilities were used to derive BCH weights reflecting each individual's uncertainty in class membership. In Step 3, the BCH weights were applied to examine differences across latent profiles on distal outcome variables. We did not include covariates in the distal outcome analyses to preserve the descriptive and person-centered nature of the profiles and to avoid over-controlling for variables that may be intrinsically related to profile membership. Given the exploratory aims, profile differences were evaluated primarily using effect sizes (odds ratios and mean differences with 95% confidence intervals), and no formal correction for multiple comparisons was applied. For all significance testing, we applied a two-tailed $\alpha = .05$ threshold.

Results

Model Selection

To identify profiles of disinhibition, we examined model fit for LPA solutions with two to four latent classes (see fit statistics in Table 1). Model selection indices supported a three-profile solution. The three-profile model showed lower BIC and AIC values relative to the two-profile model, acceptable entropy (0.80), and strong class separation as reflected in high average posterior probabilities for each class (range = 0.89–0.93). The BLRT and Lo-Mendell-Rubin adjusted LRT (LMR-A) both indicated that the three-profile model fit significantly better than the two-profile alternative. Importantly, the best loglikelihood value for the three-profile solution was replicated across all 100 final-stage random starts, providing strong evidence that the solution was stable. We rejected the 4-profile solution, because the LMR-A value was non-significant and one of the profiles comprised less than 5% of the sample.

Figure 1 shows how the three profiles differed on the trait and cognitive indicators used to determine profile membership. The first profile, which we labeled *Neurocognitive Disinhibition (n = 72)*, was defined by the lowest performance on indicators of working memory, inhibition, and inhibition/switching across the profiles, and average levels of trait impulsivity. The second profile, named *Low Disinhibition (n = 188)*, showed the lowest levels of trait urgency and low conscientiousness, and the highest scores on working memory and inhibition. The third profile, termed *Personality Disinhibition (n = 103)*, was characterized by the highest trait impulsivity across measures of urgency, low conscientiousness, and sensation seeking as well as higher scores on cognitive function than the *Neurocognitive Disinhibition* profile. Delay discounting was the only indicator that did

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not load significantly onto any profile, suggesting it was not important for distinguishing among the profiles.

As shown in Supplementary Tables S1 and S2, the final set of LPA indicators demonstrated modest-to-moderate correlations, consistent with prior research on disinhibition-related constructs, indicating that they captured related yet nonredundant dimensions of personality and cognitive functioning.

Characteristics of the Disinhibition Profiles

Lifetime Psychopathology. To test the central hypothesis that the disinhibition profiles would differ in rates of lifetime externalizing disorders, we used the BCH three-step method to test for profile differences in diagnostic status (see Table 2). This analysis indicated that the prevalence rates of externalizing disorders differed across the three profiles. As expected, the profiles marked by relatively higher scores on indicators of disinhibition, specifically the Personality Disinhibition (84.6%) and Neurocognitive Disinhibition (77.5%) profiles, showed the highest rates of externalizing disorders, with most participants meeting criteria. In contrast, the Low Disinhibition profile showed the lowest rate of externalizing disorders, though a substantial minority still meet criteria (39.9%). Pairwise BCH comparisons indicated that both the Personality and Neurocognitive Disinhibition profiles were significantly more likely than the Low Disinhibition profile to meet criteria for an externalizing disorder. However, the two disinhibition profiles did not significantly differ from each other.

To further characterize the composition of externalizing disorders within each profile, we examined profile differences in specific diagnoses. Figure 2 depicts the prevalence rates of the different externalizing disorders assessed across profiles and pairwise comparisons. Compared to the Low Disinhibition profile, the Neurocognitive and Personality Disinhibition profiles reported higher rates of alcohol use disorder, substance use disorder, and borderline personality disorder. The two profiles elevated on indicators of disinhibition did show some specificity as well, with the Personality profile showing higher rates of ADHD than the Neurocognitive profile.

We also explored whether the profiles differed in their likelihood of internalizing disorders and comorbid externalizing-internalizing disorders. Pairwise BCH comparisons revealed that individuals in the Personality Disinhibition profile (73.8%) were more likely to have an internalizing disorder compared to both the Neurocognitive Disinhibition (46.0%) and Low Disinhibition (36.0%) profiles (see Table S3 for a breakdown of specific internalizing disorders by profile). The results for the comorbid externalizing-internalizing presentation showed a similar pattern. The comorbid presentation was highest in the Personality Disinhibition profile (67.2%), followed by the Neurocognitive Disinhibition profile (42.2%) and the Low Disinhibition profile (26.7%). Although both profiles with elevations on indicators of disinhibition showed significantly higher comorbidity rates than the Low Disinhibition profile, the Personality Disinhibition profile exhibited a higher likelihood of comorbid disorders than the Neurocognitive Disinhibition profile. In summary, examination of the psychopathology correlates of the profiles suggest that the Neurocognitive Disinhibition profile has greater specificity for externalizing psychopathology, whereas the

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Personality Disinhibition profile was characterized by elevations in both externalizing and internalizing diagnoses.

History of Self- and Other-Directed Violence. Given their clinical significance, we also explored whether self-directed and other-directed violence differed across the disinhibition profiles (see Table 2). In terms of self-directed violence, the Neurocognitive Disinhibition and Personality Disinhibition profiles reported a higher average number of lifetime suicide attempts than the Low Disinhibition profile, but they did not differ from each other in terms of frequency of suicide attempts. Lifetime frequency of NSSI showed a slightly different pattern, with the Personality Disinhibition profile exhibiting significantly higher average NSSI than the Neurocognitive and Low Disinhibition profiles.

In terms of aggression against others, individuals in the Neurocognitive Disinhibition profile reported more lifetime physical aggression than the Personality Disinhibition and Low Disinhibition profiles. The Personality Disinhibition profile reported more physical aggression than the Low Disinhibition profile. Together, these findings suggest that although both high disinhibition profiles are associated with higher rates of suicide attempts, the Personality Disinhibition profile is more strongly characterized by NSSI, whereas the Neurocognitive Disinhibition profile exhibits a more pronounced pattern of other-directed violent behavior.

Trauma Exposure. Next, we examined whether the disinhibition profiles differed in lifetime exposure to traumatic events, distinguishing between assaultive and non-assaultive trauma. Profiles differed in the number of trauma types experienced across both domains. The Neurocognitive Disinhibition profile reported the greatest exposure to both assaultive and non-assaultive trauma types, followed by the Personality Disinhibition profile, with the Low Disinhibition profile reporting the lowest exposure. For assaultive trauma, pairwise BCH comparisons indicated that the Neurocognitive Disinhibition profile experienced significantly more types of assaultive trauma than both the Personality Disinhibition and Low Disinhibition profiles, and the Personality Disinhibition profile reported greater assaultive trauma exposure than the Low Disinhibition profile. A similar pattern emerged for non-assaultive trauma: individuals in the Neurocognitive Disinhibition profile experienced more non-assaultive trauma types than those in the Personality and Low Disinhibition profiles, which did not significantly differ from each other. Together, these findings indicate that profiles marked by disinhibition are characterized by elevated lifetime trauma exposure, with the Neurocognitive Disinhibition profile showing particularly high levels across both assaultive and non-assaultive trauma.

Demographic Characteristics. Finally, to aid in the interpretation of the clinical findings, we evaluated the profiles for differences in their sociodemographic characteristics. Participants in the Neurocognitive Disinhibition profile were older on average (*M/SE* = 37.47/1.41), than the Personality Disinhibition profile (*M/SE* = 30.01/1.05; Mean Difference = 7.46, SE = 1.81, 95%CI = [3.90, 11.02), and the Low Disinhibition profile (*M/SE* = 32.45/0.85; Mean Difference = 5.02, SE = 1.68, 95%CI = [1.72, 8.31]). In terms of biological sex, there were no differences between the profiles (Neurocognitive vs. Low Disinhibition: OR = 1.18 SE = 0.37, 95%CI = [0.64, 2.19]; Neurocognitive vs.

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Personality Disinhibition: OR = 0.73 SE = 0.26, 95%CI = [0.36, 1.48]; Personality vs. Low Disinhibition: OR = 0.62 SE = 0.18, 95%CI = [0.35, 1.10]). Approximately half (50.3%) of participants in the Neurocognitive profile reported female at birth, 54.5% in the Low Disinhibition profile reported female at birth and 42.5% of the participants in the Personality profile reported female at birth.

Discussion

A central challenge in the study of externalizing psychopathology is explaining the marked heterogeneity in disinhibitory processes that cut across diagnostic boundaries. The present study addressed this problem by applying a person-centered analytic approach to a multilevel battery of transdiagnostic indicators, identifying distinct patterns of disinhibition that potentially help clarify the diverse pathways associated with externalizing clinical problems. Three distinct profiles emerged from our analysis, differing in the severity and combination of disinhibitory markers as well as in their associated clinical presentations. First, the Neurocognitive Disinhibition profile was characterized by low working memory and inhibition, but unexpectedly average levels of impulsive personality traits. Individuals in this profile showed high rates of externalizing disorders and elevated engagement in other-directed violence, despite more modest levels of comorbid internalizing psychopathology. The second profile, Low Disinhibition, scored relatively low on trait impulsivity and demonstrated strong cognitive control and working memory performance, and was associated with the lowest prevalence of both externalizing and internalizing disorders, as well as minimal engagement in self- or other-directed violence. The third profile, termed Personality Disinhibition, displayed high levels of impulsive personality traits in the context of intact neurocognitive functioning and was associated with high rates of externalizing disorders, substantially elevated internalizing psychopathology, and the greatest prevalence of comorbid externalizing–internalizing conditions. Notably, this profile also showed higher engagement in non-suicidal self-injury, with comparatively lower levels of physical aggression than the Neurocognitive Disinhibition profile. Together, these findings challenge the notion of pathological disinhibition as a unitary construct, suggesting instead that different indicators of disinhibition, such as impulsive personality traits and deficits in cognitive control, may not function interchangeably or contribute equally to externalizing behavior across individuals.

A central contribution of the present study is the demonstration that externalizing psychopathology might arise through distinct underlying mechanisms, as evidenced by the divergence between the Neurocognitive and Personality Disinhibition profiles. Although both groups showed elevated externalizing psychopathology, the mechanisms related to these clinical problems appeared fundamentally different. The Neurocognitive Disinhibition profile aligned with a cognitive vulnerability model of disinhibition, such that individuals in this group exhibited impaired executive functioning, diminished inhibitory control, and high levels of trauma exposure, which are all factors known to compromise self-regulation and increase risk for externalizing behaviors (e.g., Moffitt & Caspi, 2001; Wills & Stoolmiller, 2002). This profile may reflect a high-adversity pathway, in which repeated exposure to trauma and socioeconomic hardship disrupts neurocognitive systems and may accelerate cognitive decline (Langevin et al., 2022). However, longitudinal research is

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needed to examine the hypothesis that the Neurocognitive Disinhibition profile represents a particularly entrenched and severe form of disinhibition, possibly shaped by the cumulative effects of adversity and neurocognitive disruption.

In contrast, the Personality Disinhibition profile reveals a potentially distinct, personality-based pathway to disinhibition that was marked by high levels of impulsive personality traits, but intact executive function. This finding challenges models that conceptualize externalizing psychopathology as stemming at least in part from deficits in cognitive control. The Personality Disinhibition profile appears to represent a higher-functioning subgroup of individuals with externalizing tendencies, whose impulsivity is more reflective of stable personality traits than of cognitive impairment. Although this profile showed higher rates of ADHD than the Neurocognitive and Low Disinhibition profiles, and ADHD is commonly associated with deficits in inhibitory control, sustained attention, and working memory (e.g., Loyer Carbonneau et al., 2021), these impairments were not evident in the Personality Disinhibition profile. These findings suggest that elevated ADHD in this group may reflect trait-level impulsivity rather than executive dysfunction. It is also possible that ADHD-related cognitive vulnerabilities in this subgroup are highly context dependent, emerging primarily under conditions of elevated stress, emotional arousal, or reward salience that are not fully captured by structured laboratory tasks. Moreover, individuals in this profile may possess the capacity for behavioral regulation but may be less likely to deploy inhibitory control when it conflicts with salient goals or immediate rewards. Thus, disinhibition in this group may reflect context-dependent engagement of control processes rather than uniformly diminished regulatory ability, underscoring the need for future work that integrates cognitive capacity, motivation, and ecologically valid stress or reward manipulations. Importantly, the disconnect between trait-based and neurobiological indicators of disinhibition in this group suggests that relying solely on self-report measures of trait impulsivity may obscure meaningful heterogeneity among externalizing individuals.

One intriguing possibility raised by these findings is that the Personality Disinhibition and Neurocognitive Disinhibition profiles may reflect different points along a broader developmental continuum of disinhibition rather than entirely independent subtypes. Although both profiles were marked by elevated externalizing psychopathology, individuals in the Personality Disinhibition profile were significantly younger and demonstrated intact executive functioning alongside relatively lower levels of trauma exposure, whereas individuals in the Neurocognitive Disinhibition profile were older and exhibited low cognitive performance and greater cumulative trauma. While the cross-sectional design precludes inferences about developmental progression, this pattern is consistent with the hypothesis that accumulating adversity may be associated with increasing neurocognitive dysregulation over time. This interpretation aligns with longitudinal research linking chronic psychopathology to worsening executive functioning during development (e.g., Brieant et al., 2022) and with evidence of accelerated cognitive aging and neurodegenerative processes in adulthood (Wertz et al., 2021). It is also consistent with findings that trauma exposure confers risk for both continued victimization and stress-related cognitive deterioration across the lifespan (Finkelhor et al., 2007; Miller & Sadeh, 2014). Importantly, however, the present data cannot determine whether the Personality and Neurocognitive Disinhibition profiles represent stable subtypes, developmentally linked stages, or partially overlapping

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pathways. Longitudinal studies will be essential for clarifying the temporal dynamics of these profiles and identifying potential windows for intervention to prevent worsening externalizing outcomes. Additionally, the present study did not include indicators of antagonistic externalizing in the LPA, a core dimension in trait-based models such as HiTOP (Krueger et al., 2021). Consequently, the current profiles may underrepresent externalizing pathways marked by interpersonal aggression or callous-unemotional traits. Incorporating antagonism-related indicators in future person-centered analyses may identify additional subtypes, such as aggressive or callous profiles, that are distinct from disinhibition-based pathways yet similarly elevated on externalizing risk.

In addition to the two disinhibited profiles, a third profile, termed Low Disinhibition, also emerged. This group was marked by uniformly low disinhibition across neurocognitive and personality indicators, consistent with a distinct, relatively low-risk phenotype shaped by protective factors in these domains. The pattern of findings suggests that stronger executive control, lower trait impulsivity, and more effective self-regulation, potentially in combination with reduced exposure to environmental stressors, may buffer against the expression of disinhibited behavior in this profile. As such, the Low Disinhibition profile provides a valuable comparison point for clarifying what differentiates higher-risk pathways and may inform preventive efforts focused on strengthening regulatory capacities and reducing trauma exposure in vulnerable populations. Notably, however, a subset of individuals in the Low Disinhibition group still met criteria for externalizing and internalizing psychopathology, albeit at lower rates than the more disinhibited profiles. This finding suggests that unmeasured transdiagnostic factors may confer risk through alternative mechanisms not captured by the current indicators. For example, prior work has identified subgroups characterized by high inhibitory control but low reward sensitivity, consistent with an overcontrolled pattern associated with internalizing symptoms such as anhedonia and low mood (Verdejo-Garcia et al., 2024). While such mechanisms may help explain internalizing vulnerability in this profile, they do not readily account for the presence of externalizing disorders, which may reflect additional sources of heterogeneity. Together, these findings underscore the need for future multilevel, transdiagnostic research that incorporates risk processes more specific to internalizing psychopathology to further refine low-disinhibition profiles and clarify pathways to psychopathology.

An additional issue relevant to interpretation of the profiles concerns the role of internalizing psychopathology when it co-occurs with externalizing problems. Prior literature has proposed competing hypotheses regarding this comorbidity. On one hand, the accumulation of internalizing and externalizing symptoms may reflect a more severe and complex clinical presentation. On the other hand, internalizing features, particularly anxiety-related symptoms, may exert an inhibitory influence on overt disinhibited behavior, potentially constraining impulsive or outwardly aggressive acts. In the present study, these perspectives appear to differentially apply across profiles. Individuals in the Personality Disinhibition group exhibited the highest prevalence of internalizing disorders and internalizing-externalizing comorbidity, consistent with the compounding-risk hypothesis. This group also demonstrated the greatest lifetime frequency NSSI, suggesting that internalizing pathology may amplify risk for self-directed harm. At the same time, levels of physical aggression were lower in the Personality Disinhibition profile relative to the Neurocognitive

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Disinhibition group, indicating that internalizing features may partially inhibit outwardly aggressive or confrontational behaviors while simultaneously increasing vulnerability to internalized, self-directed expressions of disinhibition.

In contrast, the Neurocognitive Disinhibition profile was characterized by high externalizing psychopathology and comparatively lower internalizing comorbidity, a pattern consistent with reduced behavioral inhibition and greater propensity toward externally directed aggression. Notably, the Personality and Disinhibition profiles did not differ in lifetime frequency of suicide attempts, although both profiles reported higher levels of these harmful behaviors that the Low Disinhibition profile. Consistent with this finding, the Low Disinhibition group showed the lowest prevalence of both internalizing and externalizing disorders, as well as comorbidity, consistent with a more resilient presentation. Together, these findings suggest that internalizing–externalizing comorbidity may not uniformly increase or decrease risk for disinhibition and violence perpetration, but instead influence the expression of disinhibition, shifting behavior toward self-directed versus other-directed harm depending on the profile.

Results also underscore the potential importance of individualized intervention strategies based on different neuropsychosocial profiles. Cognitive remediation (e.g., Baskin-Sommers et al., 2015; Klingberg, 2010) may be especially beneficial for individuals with the Neurocognitive Disinhibition profile, given prominent disruptions in executive functioning and cognitive control that likely contribute directly to impulsive and risk behaviors. In contrast, the Personality Disinhibition profile was characterized by elevated internalizing-externalizing comorbidity, emotional dysregulation, and greater self-directed harm, suggesting that disinhibition in this subgroup may be driven more by affective and motivational processes than by primary cognitive control differences. Accordingly, personality- or emotion-focused interventions that target emotion regulation and distress tolerance (e.g., Dialectical Behavior Therapy; Ciesinski et al., 2022) may be better suited to addressing risk in this group.

Interpretation of the present findings should consider the conceptual and methodological overlap between the neurocognitive and personality indicators included in the LPA. Executive control processes assessed via behavioral tasks are theoretically and empirically linked to dispositional traits such as impulsivity, negative urgency, and self-regulatory capacity (e.g, Finn, 2002; Friedman et al., 2020; Freis et al., 2022). However, these domains are measured using different modalities in the present study (performance-based tasks vs self-report, respectively), which may introduce method-related variance and contribute to weaker associations between the neurocognitive and personality indicators. Consistent with this possibility, correlations between neurocognitive and personality measures were lower than might be expected based on their theorized linkage (see Supplemental Tables S1 & S2). This pattern likely reflects multiple processes. First, cognitive control abilities and impulsive personality traits, while related, represent partially distinct constructs that operate at different levels of analysis, capturing momentary performance capacity versus stable dispositional tendencies, respectively (Friedman et al., 2020). Second, method variance associated with task-based versus self-report measures may attenuate observed correlations, obscuring shared variance that could be more evident in multimethod or latent-variable

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approaches (e.g., Venables et al., 2018). Finally, differences in context, for example neutral task contexts versus emotion-based self-report items, may further contribute to dissociations across domains. Future research would benefit from research designs that more directly disentangle shared and distinct sources of variance across neurocognitive and personality domains. Notably, because LPA is designed to identify patterns of observed scores, method-related variance may have influenced our profile formation, which is a limitation of this study. Multimethod assessment strategies, such as the use of both task-based and self-report measures for assessing executive function and personality traits where feasible, are needed to clarify whether the observed profiles primarily reflect true process-level heterogeneity or method-specific effects. Such work will be critical for the conceptual replication of these findings and for advancing etiological models of disinhibition.

Limitations to the study design should be acknowledged to contextualize the findings. First, it is possible that additional disinhibition profiles were not detected due to limited statistical power. Furthermore, the stability of the identified profiles is uncertain, as the cross-sectional design and absence of a replication sample prevent conclusions about their robustness or how profile membership may change over time. Additionally, the cross-sectional nature of the study precludes inferences about causality or the development of disinhibition-related psychopathology. Finally, internalizing psychopathology was examined at a broad level rather than being disaggregated into more specific diagnostic groupings (e.g., mood vs. anxiety disorders). This decision was made to limit multiple testing but may have obscured more nuanced patterns of comorbidity.

At the same time, the study design has several notable strengths to highlight. We incorporated behavioral, cognitive, interview, and self-report measures to capture disinhibition across multiple levels of analysis. The sample was diverse in terms of race/ethnicity, socioeconomic status, and exposure to adversity, enhancing the generalizability of the findings. Additionally, despite being a community-dwelling sample, participants showed elevated rates of psychopathology, allowing for meaningful examination of clinical phenomena outside of institutional or treatment settings.

In conclusion, this study highlights meaningful heterogeneity among disinhibited individuals by identifying distinct neuropsychosocial profiles using a person-centered, transdiagnostic approach. Through latent profile analysis, three distinct subtypes emerged, each reflecting unique constellations of personality traits and neurocognitive functioning. Importantly, two high-disinhibition profiles were both associated with elevated externalizing psychopathology and violence perpetration, despite differing markedly in personality and cognitive functioning. These findings underscore the value of symptom-agnostic, multilevel classification frameworks in capturing the complexity of disinhibition and externalizing psychopathology.

Supplementary Material

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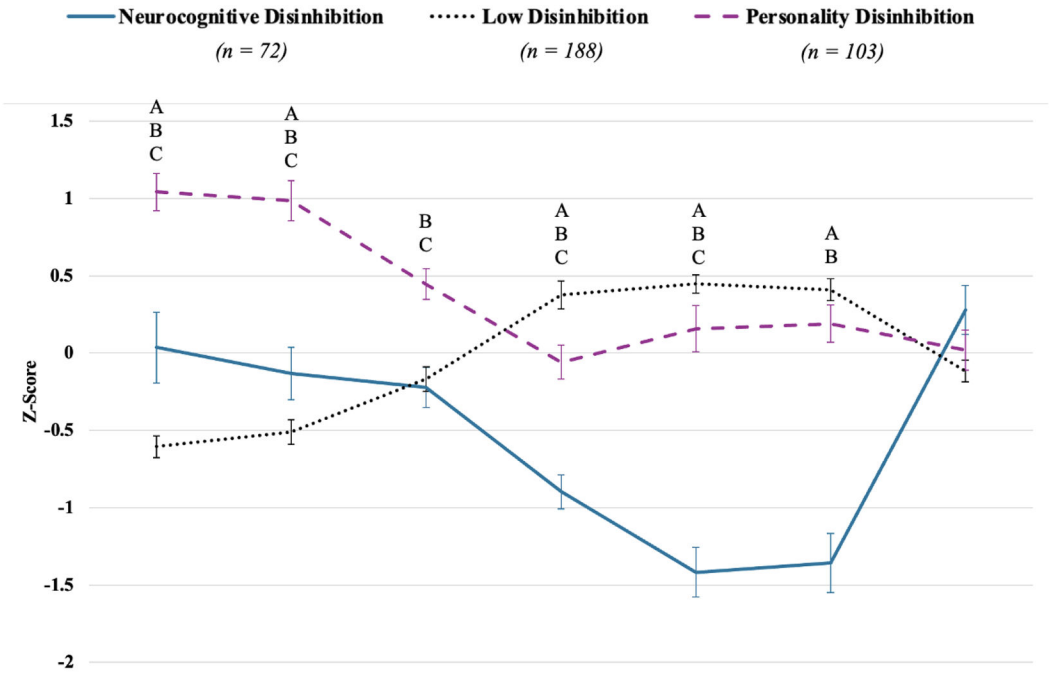

Profile	Urgency	Low Conscientiousness	Sensation Seeking	Digit Span	Inhibition	Inhibition/ Switching	Delay Discounting
Neurocognitive Disinhibition	0.04/.24	-0.13/.17	-0.22/.13	-0.90/.11*	-1.42/.16*	-1.36/.19*	0.28/.16
Low Disinhibition	-0.61/.07*	-0.51/.08*	-0.17/.08	0.38/.09*	0.45/.06*	0.41/.07*	-0.12/.07
Personality Disinhibition	1.04/.12*	0.99/.13*	0.45/.10*	-0.06/.11	0.16/.15	0.19/.12	0.02/.13

Figure 1.

Disinhibition Profiles

Note. Error bars reflect standard errors of the mean. Means and standard deviations for each indicator within each profile are presented in the accompanying table. Asterisks next to table values indicate indicators that significantly load on the corresponding profile at $p<.001$. One-way ANOVAs with three profiles as the factor and the indicators as the dependent variables were all significant at $p < .001$ [$Fs_{(2, 361)} = 17.70–204.45$, $\eta^2 = .09–.53$] except for delay discounting [$F_{(2, 361)} = 3.76$, $\eta^2 = .02$, $p = .023$]. Post hoc contrasts using the Games–Howell correction for multiple comparisons showed the following significant pairwise differences at $p < .001$: **A** = Neurocognitive Disinhibition vs. Low Disinhibition; **B** = Neurocognitive Disinhibition vs. Personality Disinhibition; **C** = Low Disinhibition vs. Personality Disinhibition.

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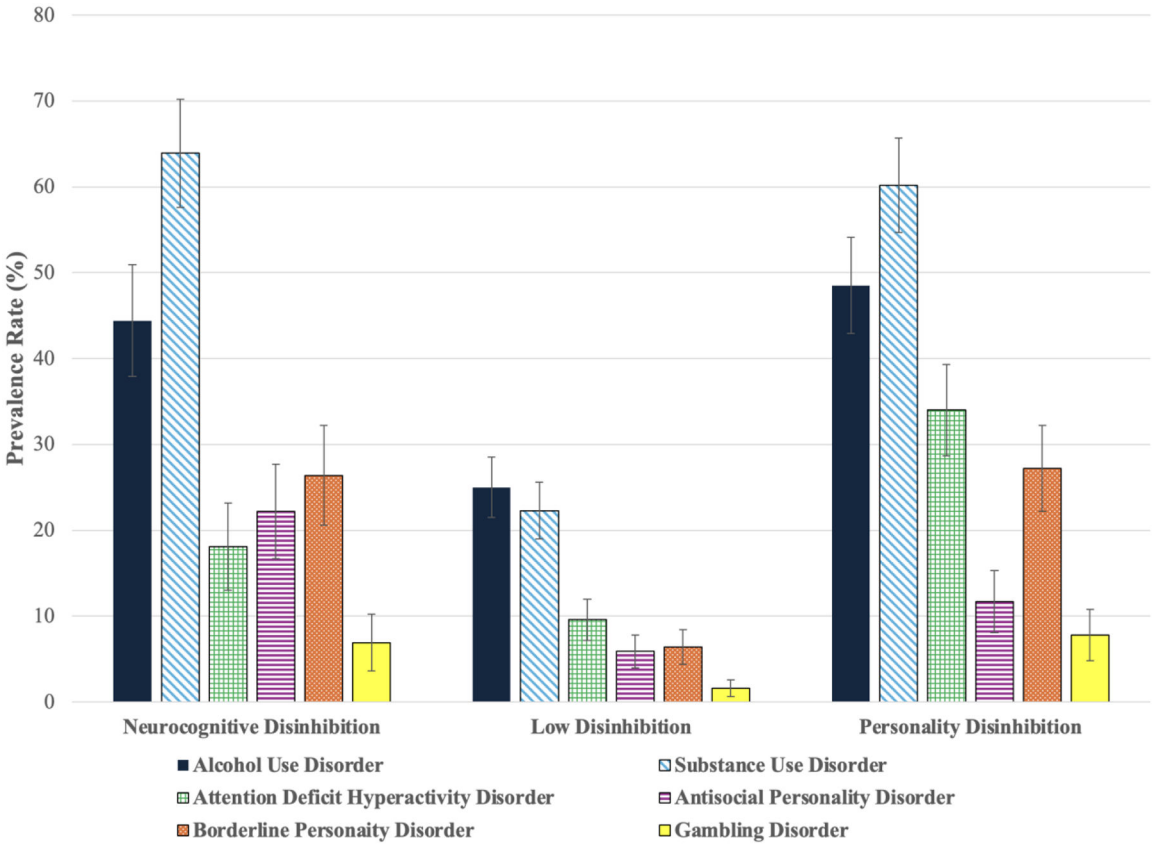

	Neurocognitive vs. Low	*Personality vs. Low*	*Neurocognitive vs. Personality*
	OR/ 95% CI	*OR/ 95% CI*	*OR/ 95% CI*
Alcohol Use Disorder	**2.74 [1.42, 5.29]**	**3.50 [1.88, 6.54]**	0.78 [0.39, 1.58]
Substance Use Disorder	**8.28 [4.06,16.91]**	**7.63 [3.88, 14.93]**	1.09 [0.51, 2.30]
Attention Deficit Hyperactivity Disorder	2.49 [0.94, 6.57]	**6.94 [2.99, 16.13]**	**0.36 [0.15, 0.84]**
Antisocial Personality Disorder	**5.75 [2.13, 15.53]**	2.53 [0.85, 7.46]	2.28 [0.90, 5.76]
Borderline Personality Disorder	**7.81 [2.63, 23.14]**	**8.70 [2.96, 25.64]**	0.90 [0.42, 1.94]
Gambling Disorder	7.09 [0.80, 62.76]	8.55 [0.98, 76.93]	0.83 [0.23, 3.04]

Figure 2.

Lifetime Externalizing Diagnoses by Disinhibition Profile

Note. Percentage of individuals who met lifetime criteria for an externalizing disorder. Bars indicate +/−1 standard error. The statistics below the bar graph reflect BCH pairwise comparisons between the profiles. Bolded text indicates differences.

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Table 1.

Fit Statistics for Disinhibited Profiles (N = 363)

Model	AIC	BIC	Entropy	BLRT *p*	LMR-A *p*
2 Profiles	6843.87	6929.54	0.86	<.001	<.001
3 Profiles	6673.08	6789.91	0.80	<.001	.011
4 Profiles[+]	6610.66	6758.64	0.84	<.001	.162

Note. AIC = Akaike Information Criterion. BIC = Bayesian Information Criterion. BLRT = bootstrapped likelihood ratio test. LMR-A = Lo-Mendell-Rubin–Adjusted likelihood ratio test P-value.

[+]Profile rejected due because one profile was <3% of the sample.

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Table 2.

Clinical Characteristics of the Disinhibition Profiles

	Neurocognitive Disinhibition (n = 72)	Low Disinhibition (n = 188)	Personality Disinhibition (n = 103)	Pairwise Comparisons		
				Neurocognitive vs. Low	Personality vs. Low	Neurocognitive vs. Personality
Lifetime Psychopathology	*PR*	*PR*	*PR*	*OR [95% CI]*	*OR [95% CI]*	*OR [95% CI]*
Externalizing Disorder	77.5%	39.9%	84.6%	**5.20 [2.53, 10.67]**	**8.26 [3.79, 18.18]**	0.63 [0.24, 1.63]
Internalizing Disorder	46.0%	36.0%	73.8%	1.52 [0.81, 2.83]	**5.03 [2.60, 9.71]**	**0.30 [0.14, 0.65]**
Comorbid EXT-INT Disorder	42.2%	26.7%	67.2%	**2.01 [1.05, 3.85]**	**5.62 [2.95, 10.75]**	**0.36 [0.17, 0.75]**
Self-/Other- Directed Violence	*M/SE*	*M/SE*	*M/SE*	*MD [95% CI]*	*MD [95% CI]*	*MD [95% CI]*
Suicide Attempts	0.23/0.14	−0.25/0.05	0.27/0.14	**0.48 [0.18, 0.77]**	**0.52 [0.21, 0.82]**	−0.04 [−0.44, 0.36]
Non-Suicidal Self-Injury	−0.03/0.11	−0.23/0.05	0.42/0.15	0.19 [−0.04, 0.42]	**0.64 [0.31, 0.97]**	**−0.45 [−0.83, −0.07]**
Physical Aggression	0.56/0.14	−0.30/0.06	0.15/0.10	**0.86 [0.56, 1.16]**	**0.46 [0.22, 0.69]**	**0.40 [0.06, 0.74]**
Trauma Exposure	*M/SE*	*M/SE*	*M/SE*	*MD [95% CI]*	*MD [95% CI]*	*MD [95% CI]*
Assaultive	3.91/0.26	1.97/0.16	3.01/0.24	**1.94 [1.33, 2.55]**	**1.04 [0.44, 1.64]**	**0.90 [0.19, 1.61]**
Non-Assaultive	1.14/0.12	0.67/0.07	0.79/0.09	**0.47 [0.20, 0.74]**	0.12 [−0.11, 0.35]	**0.35 [0.05, 0.65]**

N = 363. Bolded text indicates differences. EXT = Externalizing. INT= Internalizing. PR = Prevalence Rate. OR = Odds Ratio. CI = Confidence interval. MD = Mean difference. Psychopathology rates reflect lifetime diagnoses. Mean levels of suicide attempts, non-suicidal self-injury, and physical aggression represent lifetime estimates. Trauma exposure reflects average lifetime exposure to different types of traumatic events.